Answer Key & Teacher Notes

Session 1 — Using Properties to Rewrite Expressions · Session 2 — Expanding and Factoring Expressions

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Using Properties to Rewrite Expressions

TALK PROMPTS

- Why does rewriting an expression with properties never change the value it gives?
- Between your two answers you used a property to move terms around. Which one, and why is it allowed?

PARTNER TASK

Two Steps Together. Partner A distributes in 5(j + 6) + 4j; Partner B combines like terms and names the property used to reorder. Both check with j = 1 (39 and 39). Switch roles with 2(k + 10) + 4k.

WATCH FOR

- Multiplying only the first term inside the parentheses (4(q + 5) written as 4q + 5).
- Combining unlike terms (7m + 8 written as 15m).
- Using the Commutative Property for subtraction or division.

SESSION 2 · TEACHER NOTES

Expanding and Factoring Expressions

TALK PROMPTS

- How are expanding and factoring opposite moves? Explain what each one does.
- One answer removed parentheses and the other added them. How can both moves keep the value the same?

PARTNER TASK

Both Forms. Partner A expands 6(y + 7); Partner B factors 14b + 35 with the GCF. Trade papers and check each answer by doing the opposite move.

WATCH FOR

- Factoring out a common factor that is not the greatest (16y + 40 = 2(8y + 20)).
- Multiplying only the first term when expanding (5(w + 8) = 5w + 8).
- Dividing only one term by the GCF when factoring (18x + 24 = 6(18x + 4)).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.A.3	1	1	2	1	1	2	3	2	2	1	1
S1 · Form B	6.EE.A.3	1	1	2	1	1	2	3	2	2	1	1
S2 · Form A	6.EE.A.3	1	1	2	1	1	2	2	3	2	1	1
S2 · Form B	6.EE.A.3	1	1	2	1	1	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

Distributive Property: Multiply the outside number by **each** term inside the parentheses.

Commutative Property: Changing the **order** when adding or multiplying does not change the answer.

Associative Property: Changing the **grouping** when adding or multiplying does not change the answer.

like terms: Terms with the same **variable** part.

- Expressions are **equivalent** when they give the same value for every value of the variable.
- To simplify: distribute, put like terms together, then **combine** them.

Example: Distribute: 4(m + 2) = 4m + **8**. Reorder: 4m + 3m + 8 (Commutative Property). Combine like terms: 4m + 3m = **7m**. Answer: 4(m + 2) + 3m = **7m + 8**

YOU TRY

1. a. 2x + 10 b. 5y + 3

2. 7c + 16; check: 6(3) + 1 + 4 = 23 and 7 + 16 = 23

EXIT TICKET

1. a. 5x + 15 b. 8n + 8

2. 7b + 8

3. **B** (Associative)

PRACTICE FORM A

1. a. 4n + 20 b. 8x + 6 c. 15y + 18 d. 5a + 11

2. a. Associative b. Commutative c. Identity

3. a. 3n + 12 b. 8n + 12 c. 28 and 28

4. 7p + 23

5. 10q + 19

6. No. 24 ÷ 6 = 4 but 6 ÷ 24 = 1/4. The Commutative Property works for adding and multiplying, not dividing.

7. Sample: 3w + 6 + w (Distributive); 3w + w + 6 (Commutative); 4w + 6 (combine like terms)

8. a. 3a + 18 and 2a + 8 b. 5a + 26 c. 51 tickets

9. The 4 must multiply every term inside, so 4 × 5 = 20 too. 4(q + 5) = 4q + 20.

10. **B** (5*x* + 20)

11. **B** (7*n* + 6)

Self-check word **PROPERTY**

PRACTICE FORM B

1. a. 6n + 42 b. 11x + 12 c. 13y + 10 d. 5a + 16

2. a. Associative b. Commutative c. Identity

3. a. 4n + 12 b. 10n + 12 c. 42 and 42

4. 9p + 16

5. 10q + 17

6. No. 30 ÷ 5 = 6 but 5 ÷ 30 = 1/6. The Commutative Property works for adding and multiplying, not dividing.

7. Sample: 2w + 10 + 3w (Distributive); 2w + 3w + 10 (Commutative); 5w + 10 (combine like terms)

8. a. 4a + 12 and 2a + 10 b. 6a + 22 c. 46 tickets

9. The 3 must multiply every term inside, so 3 × 7 = 21 too. 3(k + 7) = 3k + 21.

10. **C** (7*b* + 42)

11. **B** (every term inside)

Self-check word **IDENTITY**

GUIDED NOTES

expand: Multiply to **remove** the parentheses.

factor an expression: Write it as a **product**: a shared factor times a sum.

term: One **part** of an expression, separated by + or − signs.

greatest common factor: The **biggest** number that divides both numbers evenly.

- Factoring **undoes** expanding, so you can check a factored form by expanding it.
- Factor out the **GCF** so the terms inside share no other common factor.

Example: The GCF of 18 and 27 is **9**. Divide each term: 18x ÷ 9 = **2x** and 27 ÷ 9 = 3. Check: 9 × 2x + 9 × 3 = 18x + 27. Answer: 18x + 27 = **9(2x + 3)**

YOU TRY

1. a. 8x + 12 b. 5(3y + 4)

2. 8(2n + 3); check: 8 × 2n + 8 × 3 = 16n + 24

EXIT TICKET

1. a. 7a + 35 b. 6(2b + 3)

2. 10(2m + 3)

3. **B** (9*y* + 18)

PRACTICE FORM A

1. a. 8c + 24 b. 10d + 20 c. 6y + 42 d. 7n + 14

2. a. 4(3m + 5) b. 5(3k + 7) c. 7(2b + 5) d. 6(3x + 4)

3. a. 4t + 36 b. 5 c. 5(4t + 9)

4. 5x + 30

5. 9z + 24

6. No; 8y and 20 still share 4. GCF form: 8(2y + 5)

7. 6(5c + 7); check: 6 × 5c + 6 × 7 = 30c + 42

8. a. 12(2p + 3) b. 12 equal rows with 2p + 3 plants in each row c. 24(4) + 36 = 132 and 12(11) = 132

9. Lucia forgot to multiply 8 by 5. The 5 multiplies every term inside: 5(w + 8) = 5w + 40.

10. **B** (5(2*x* + 3))

11. **B** (6*y* + 24)

Self-check word **FACTORS**

PRACTICE FORM B

1. a. 9w + 36 b. 12d + 15 c. 5y + 45 d. 8n + 24

2. a. 6(4q + 3) b. 7(3k + 4) c. 5(2b + 5) d. 6(2x + 5)

3. a. 6s + 48 b. 7 c. 7(4s + 7)

4. 6x + 18

5. 8z + 15

6. No; 6y and 15 still share 3. GCF form: 9(2y + 5)

7. 9(a + 3); check: 9 × a + 9 × 3 = 9a + 27

8. a. 6(3p + 5) b. 6 equal rows with 3p + 5 plants in each row c. 18(4) + 30 = 102 and 6(17) = 102

9. Amina forgot to multiply 3 by 7. The 7 multiplies every term inside: 7(k + 3) = 7k + 21.

10. **B** (4(2*n* + 3))

11. **A** (greatest common factor)

Self-check word **PRODUCT**

SHEET 1 · USING PROPERTIES TO REWRITE EXPRESSIONS

1. a. $6n + 12$ b. $12x + 3$ c. $10 + 5y$ d. $7a - 21$ e. $6b + 10$ f. $10m - 10$

2. a. $6(30 + 4) = 180 + 24 = 204$ b. $8(50 + 2) = 400 + 16 = 416$

3. a. $9c + 12$ b. $3x + 13$ c. $7m + 10$ d. $5t + 18$

4. $6x + 24$; when $x = 5$: $6 \times 5 + 24 = 54$ ft² (check: $6 \times 9 = 54$)

5. She did not multiply the 4 by 2. $2(3x + 4) = 6x + 8$, so $6x + 8 + x = 7x + 8$.

SHEET 2 · EXPANDING AND FACTORING EXPRESSIONS

1. a. $15x + 6$ b. $8y + 56$ c. $12 + 8n$

2. a. $3(2x + 5)$ b. $4(5y + 8)$ c. $7(2a + 7)$ d. $9(4 + 3k)$ e. $20(2m + 3)$ f. $11(b + 3)$

3. $8x + 56 = 8(x + 7)$, so the length is $x + 7$ units

4. Sample: $2(6x + 12)$, $3(4x + 8)$, $4(3x + 6)$, or $6(2x + 4)$. GCF form: $12(x + 2)$.

5. No. $6n$ and 9 still share a factor of 3. The GCF is 15: $15(2n + 3)$.

6. **B** $(8(2x + 5))$

7. **C** $(18a + 12)$

October 2026 conceptual additions — teacher key

Session 1 · Using Properties to Rewrite Expressions

3x+12; then 5x+12. Distribution applies 3 to both terms; regroup like terms using properties of addition.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 3x+12; then 5x+12. Distribution applies 3 to both terms; regroup like terms using properties of addition.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Expanding and Factoring Expressions

GCF 6 gives 6(2x+3); expanding recovers 12x+18. The other expression is equivalent but uses common factor 2 instead of greatest factor 6.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: GCF 6 gives 6(2x+3); expanding recovers 12x+18. The other expression is equivalent but uses common factor 2 instead of greatest factor 6.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.